

# Serotonin-Restrained FOXO Stress Transcription Model in Major Depressive Disorder — Hypothesis-Search Report

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**Hypothesis ID:** `serotonin_foxo_stress_transcription_model` **Hypothesis Label:** Serotonin-Restrained FOXO Stress Transcription Model **Target Disease:** Major Depressive Disorder (MDD; MONDO\_0002009), Category: Complex **KB Status:** EMERGING **Report date:** 2026-10-09 (OpenScientist autonomous run, 5 iterations)

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## Executive Judgment

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**Verdict: Partially supported overall, but only weakly supported in humans.** The hypothesis proposes that serotonergic tone restrains FOXO-family stress transcription through insulin/IGF-1 receptor signalling, so that loss of serotonin de-represses a FOXO-driven stress program on which antidepressants, lithium, and clozapine act. After five iterations integrating primary literature, a negative human cohort, Mendelian-randomization evidence, and a computational interrogation of the Open Targets Platform (v26.9), the evidence splits sharply by organism.

The **model-organism backbone is strong and internally consistent:** in *Caenorhabditis elegans*, serotonin-null *tph-1* mutants phenocopy insulin-receptor *daf-2* mutants in DAF-16/FOXO nuclear accumulation; exogenous serotonin and fluoxetine prevent that accumulation; the insulin receptor sits genetically downstream of serotonin; and lithium and clozapine converge on the same DAF-16 localisation readout through  $\beta$ -arrestin and SGK ([PMID:17141627](#), [PMID:10676966](#), [PMID:21732403](#)). Independently, **mammalian stress models repeatedly implicate an AKT→FOXO node** whose nuclear/active state is pro-depressive and whose cytoplasmic sequestration is antidepressant, with fluoxetine among the reversing agents ([PMID:36819781](#), [PMID:36039087](#), [PMID:41249439](#), [PMID:37173696](#)).

The **human arm, however, does not close the loop.** No study demonstrates a serotonin→FOXO edge in any mammal; the specific causal step that defines the hypothesis is untested outside the nematode. Rodent directionality is internally contradictory (nuclear

FOXO3a is pro-depressive in one model, yet dentate-gyrus FoxO3a overexpression is protective in another). A direct human cohort found plasma FOXO3 unassociated with depressive symptoms after adjustment ([PMID:42632434](#)). Human genetics (Open Targets v26.9) nominates only FOXO1 — not the rodent-favoured FOXO3 — with a single, moderate MDD credible-set signal that traces to a *cognition*×*MDD interaction* GWAS rather than a primary MDD case/control scan, while the insulin-receptor genes (INSR, IGF1R, SGK1, AKT1) have no MDD genetic support at all, and FOXO1 ranks far below MDD's monoaminergic/synaptic genetic core (DRD2, HTR2A, GRIN2A, GABRA1). Finally, the **upstream "loss of serotonin" premise is itself contested in humans** by the Moncrieff umbrella review, surviving at best as a subtype-restricted claim. The recommended curation outcome is a **HUMAN\_MODEL\_MISMATCH discussion attached to Monoamine Deficiency**, with at most a subtype-restricted EMERGING FOXO1 lead — **not** a primary insulin/FOXO pathophysiology node.

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## Key Findings

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### F001 — The serotonin→DAF-2→DAF-16/FOXO chain is well-evidenced in *C. elegans* (seed backbone)

The hypothesis's foundational evidence is a single, coherent body of nematode genetics. In [PMID:17141627](#), serotonin-deficient *tph-1* mutants, like *daf-2* insulin-receptor mutants, exhibit DAF-16 nuclear accumulation and constitutive physiological stress states, and *"Exogenous 5HT and fluoxetine (Prozac) prevented DAF-16 nuclear accumulation in wild-type animals under stresses."* Genetic epistasis places DAF-2 downstream of 5-HT signalling, with distinct serotonergic neurons acting through distinct 5-HT receptors to influence distinct DAF-16-mediated stress responses. The drug-convergence claim is supported in [PMID:21732403](#): *"Lithium, a mood stabilizer often used to treat psychosis, also requires  $\beta$ -arrestin and SGK to suppress the nuclear localization of DAF-16."* An earlier characterisation ([PMID:10676966](#)) first tied serotonin loss to reduced insulin-like neuroendocrine signalling: *"This metabolic dysregulation is, in part, due to downregulation of transforming growth factor-beta and insulin-like neuroendocrine signals."*

This is the strongest **direct** evidence for the hypothesis as stated — but its directness for MDD is INDIRECT on two counts: the organism is a nematode, and the readout is subcellular DAF-16 localisation, not a depression phenotype. The chain is real and reproducible in worms; the question is whether it generalises to human disease.

## F002 — Mammalian stress models independently support a pro-depressive AKT→FOXO node reversed by antidepressants

Multiple independent rodent studies converge on an AKT→FOXO axis in which **nuclear/active FOXO is pro-depressive and cytoplasmic sequestration is antidepressant**:

- **PMID:36819781** — Chronic unpredictable mild stress (CUMS) produced *"increased expression of AKT, FOXO, and Bim in the hippocampus. These alterations were ameliorated by administration of 20 mg/kg or 40 mg/kg of traxoprodil"* (with fluoxetine as comparator), raising the pro-apoptotic FOXO target Bim and reducing BDNF/pERK/pCREB.
- **PMID:36039087** — The antidepressant Chaihu Shugan San increases SIRT1 and *"could also promote FOXO1 translocation to the cytoplasm through SIRT1 signaling, which triggered FOXO1 protein degradation,"* i.e., it acts by excluding/degrading FOXO1.
- **PMID:41249439** — NDRG2 upregulation inhibits AKT and *"promotes the nuclear translocation of FoxO3a, leading to the transcriptional activation of the pro-apoptotic factor Puma,"* driving astrocyte apoptosis and depressive-like behaviour.
- **PMID:37173696** and **PMID:40237232** — TLR4/PI3K/AKT/FOXO1 and SIRT1/FOXO1 antidepressant mechanisms, both acting by suppressing/excluding FOXO1.

This is the hypothesis's best **mammalian** support. Crucially, however, **none of these studies manipulates serotonin to move FOXO**; they establish the downstream AKT→FOXO→stress-program arm but leave the defining serotonin→FOXO edge untested.

## F003 — Directional conflict in mammals and a negative human biomarker

The mammalian FOXO picture is not directionally clean. Against the pro-depressive framing of F002, **PMID:41622191** shows that AAV overexpression of FoxO3a in the dentate gyrus *alleviates* CUS-induced anxiety- and depression-like behaviours and cognitive impairment — FOXO3a as **protective** (CUS *downregulates* dentate-gyrus FoxO3a, and restoring it rescues the phenotype). So within rodents FOXO3a is cast as both pro- and anti-depressive depending on cell type and region (astrocytic apoptosis vs. dentate-gyrus neuroplasticity).

Human evidence is weak-to-negative. In a cerebral small-vessel-disease cohort (183 patients vs. 150 controls), plasma FOXO3 was elevated with cognitive impairment but, *"After adjustment, FOXO3 remained unassociated with depressive symptoms ( $P = 0.232$ )"* (**PMID:42632434**). No human MDD brain or genetic study directly linking serotonergic tone to FOXO was retrieved.

## F004 — Human genetics nominates FOXO1 (not FOXO3/4), and the insulin-receptor arm has no genetic support

Interrogating the **Open Targets Platform GraphQL API (v26.9.0, disease MONDO\_0002009, retrieved 2026-10-09)** produced a clear picture:

| Gene            | Overall score  | Genetic-association datatype | GWAS credible-set rows   | Notes                         |
|-----------------|----------------|------------------------------|--------------------------|-------------------------------|
| <b>FOXO1</b>    | 0.187          | 0.285                        | 1 (L2G=0.469)            | Only FOXO with a credible set |
| FOXO3           | 0.038          | —                            | 0                        | Literature-only               |
| FOXO4 / FOXO6   | no association | —                            | 0                        | —                             |
| INSR            | 0.281          | —                            | 0                        | Clinical + literature only    |
| IGF1R           | 0.014          | —                            | 0                        | Literature only               |
| SGK1            | 0.101          | —                            | 0                        | Literature only               |
| AKT1            | 0.083          | —                            | 0                        | Literature only               |
| SLC6A4 (anchor) | 0.637          | —                            | clinical 0.999           | Positive control              |
| TPH2 (anchor)   | 0.442          | 0.375                        | genetic_literature 0.608 | Positive control              |

The serotonin genes behave as expected positive anchors, validating the query. But the FOXO-family signal is carried by **FOXO1 alone**, with only a moderate genetic association, and **every gene in the insulin/IGF-1 receptor arm lacks any MDD GWAS credible-set or gene-burden evidence**. This is a direct mismatch with the hypothesis, which centres the insulin-receptor edge and whose mammalian mechanism rests on FOXO3-type biology.

## F005 — The insulin pathway links to MDD in humans, but as a metabolic/comorbid arm, not a serotonin→FOXO edge

Human Mendelian randomization does support an insulin-pathway connection to MDD — just not the one the hypothesis draws. In [PMID:40595333](#): *"a doubling in MDD genetic liability was associated with 1.14 higher odds of T2D (95% CI:1.09, 1.19), whilst a doubling in T2D genetic liability associated with 1.02 higher odds of MDD (95% CI:1.01, 1.03),"* with an obesity-mediated insulin-resistance pathway implicated (OR 1.06, 95% CI 1.04–1.09) and BMI attenuating the relationship. This is corroborated by multi-trait GWAS ([PMID:41555281](#)) and a genomic systematic review ([PMID:41780463](#)) reporting shared MDD–metabolic-syndrome architecture and a unidirectional MDD→MetS causal effect. The insulin node in human MDD is therefore real, but it operates as a **parallel metabolic/comorbid link** rather than as the serotonin-gated transcriptional switch the hypothesis requires.

## F006 — The sole FOXO1 signal derives from a cognition×MDD interaction GWAS, not a primary MDD GWAS

Resolving the FOXO1 credible set (Open Targets v26.9, retrieved 2026-10-09) qualifies the human-genetics support substantially. The single `gwas_credible_sets` evidence is variant **rs180828263** (13\_40534758\_C\_T, chr13:40,534,758, ~21 kb 5' of FOXO1),  $p=4\times 10^{-9}$ , PICS fine-mapped *"based on reported top hit"* (not full statistical fine-mapping), L2G=0.469. Its parent study is GCST90096932 = **"Cognitive function (executive function) × major depressive disorder interaction (2df)"** GWAS ([PMID:34782712](#),  $n=9,567$ ) — a gene×phenotype interaction scan, **not** a primary MDD case/control GWAS. The one human genetic thread connecting FOXO to MDD is thus about a *cognition-conditioned* MDD phenotype, narrowing any claim to a cognitive subtype.

## F007 — The upstream "loss of serotonin" premise is contested; serotonin deficiency may characterise only a subtype

The hypothesis's upstream node assumes "loss of serotonin." The Moncrieff et al. systematic umbrella review ([PMID:35854107](#); republished [PMID:42538881](#)) *"aimed to synthesise and evaluate evidence on whether depression is associated with lowered serotonin concentration or activity"* and found no consistent evidence that depression is associated with lowered serotonin concentration or activity. A published riposte ([PMID:36356194](#)) agrees the simple equation is outdated but argues *"The decreased activity of serotonin, which undoubtedly plays an essential role in the pathogenesis of depression, is characteristic of only a subgroup of*

*depressed subjects*" (agitated/anxious, insomnia, reduced appetite, SSRI-responsive), with anhedonic/anergic cases driven more by dopamine/noradrenaline. Separately, **direct PubMed searches for a mammalian serotonin→FOXO edge returned nothing usable** (searches 2026-10-09: "serotonin 5-HT receptor FOXO transcription factor brain neurons" returned only a *C. elegans* paper; "fluoxetine FOXO3a hippocampus phosphorylation" and "fluoxetine Akt FOXO neuronal" returned 0 results).

## F008 — FOXO1 ranks far below MDD's neurotransmitter/synaptic genetic core

Contextualising within the full MDD target landscape (Open Targets v26.9, MONDO\_0002009, 6,780 associated targets, retrieved 2026-10-09), the top MDD targets by overall score are neurotransmitter/synaptic genes led by genetic-association signals: **DRD2** (0.743, genetic 0.854), **HTR2A** (0.703, 0.547), **GRIN2A** (0.675, 0.472), **GABRA1** (0.672, 0.423), followed by SLC6A4, HTR1A, SLC6A3/2, multiple GABRs, DRD3/4, MAOA, GRIN2B. **FOXO1 (0.187) does not appear in the top 30; FOXO3 (0.038) is literature-only**. The dominant human genetic architecture of MDD favours monoamine/glutamate/GABA models over a FOXO-insulin model.

## F009 — Integrated synthesis: HUMAN\_MODEL\_MISMATCH with a subtype-restricted FOXO1 lead

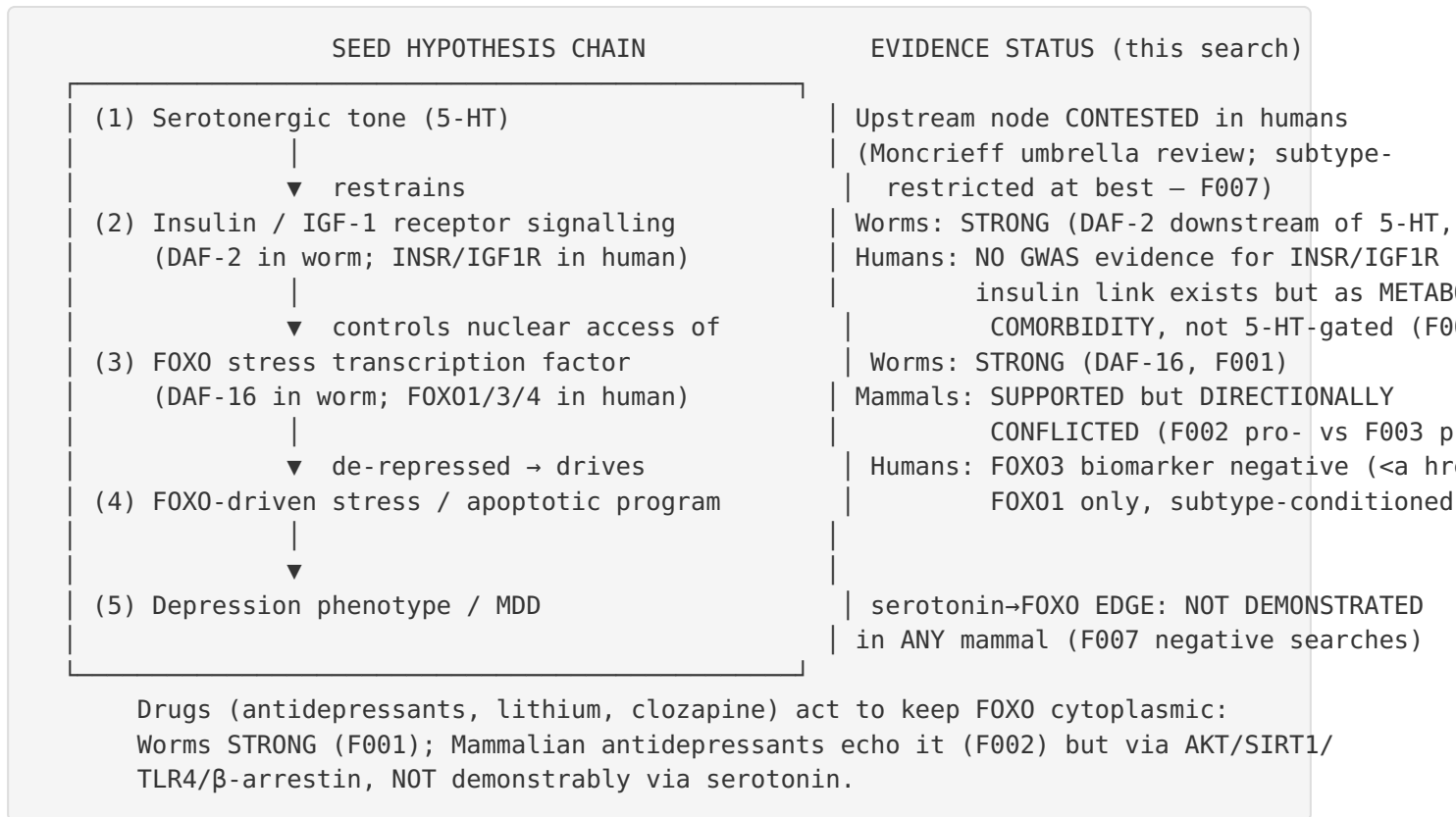
Integrating all eight findings, the hypothesis is **partially/weakly supported in humans**. A strong *C. elegans* backbone and independent mammalian AKT→FOXO support coexist with rodent directional conflict, a negative human biomarker, an insulin arm supported only as a comorbid metabolic link, a contested upstream serotonin premise, no direct mammalian serotonin→FOXO paper, and a human FOXO1 genetic signal that is both subtype-conditioned and far outside MDD's genetic core.

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## Mechanistic Model / Interpretation

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The hypothesis asserts a single causal chain. The diagram below annotates each edge with the strength of evidence found in this search.



**Interpretation.** The chain is fully instantiated and reproducible only in *C. elegans*. In mammals, the **downstream half** (AKT→FOXO→stress/apoptosis, reversible by antidepressants) is independently supported, but the **upstream half** that makes the hypothesis distinctive — serotonin *gating* FOXO via the insulin receptor — is the weakest link and is untested in any mammal. Two human-level facts actively pull against promoting a primary node: (i) the insulin-receptor genes carry no MDD genetic signal while FOXO1 (the wrong family member relative to the mammalian mechanism) carries only a subtype-conditioned one; and (ii) the human insulin→MDD relationship that *does* exist is a metabolic/obesity comorbidity, a parallel mechanism rather than the serotonin-gated switch. The most parsimonious reading is that FOXO is a **convergent downstream stress-transcription hub** engaged by many antidepressant mechanisms (SIRT1, PI3K/AKT, TLR4, β-arrestin/SGK), rather than a serotonin-specific effector.

## Evidence Base (Evidence Matrix)

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| Citation                      | Evidence type                        | Role                         | Mechanistic claim tested                            | Key finding                                                                                                          |
|-------------------------------|--------------------------------------|------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <a href="#">PMID:17141627</a> | Model organism ( <i>C. elegans</i> ) | <b>Supports</b>              | Serotonin restrains DAF-16/FOXO via DAF-2           | <i>tph-1</i> phenocopies <i>daf-2</i> ; 5-HT & fluoxetine prevent DAF-16 nuclear accumulation; DAF-2 downstream 5-HT |
| <a href="#">PMID:10676966</a> | Model organism                       | <b>Supports</b>              | Serotonin loss → reduced insulin-like signalling    | Serotonin-synthesis mutant downregulates TGF- $\beta$ and insulin neuroendocrine signals                             |
| <a href="#">PMID:21732403</a> | Model organism                       | <b>Supports</b>              | Lithium/clozapine act on FOXO localisation          | Both require $\beta$ -arrestin + SGK to block DAF-16 cytoplasmic                                                     |
| <a href="#">PMID:36819781</a> | Model organism (rodent)              | <b>Supports (downstream)</b> | Chronic stress raises hippocampal AKT/FOXO/Bim      | CUMS ↑ AKT/FOXO/Bim; reversed by traxoprodil (fluoxetine comparable)                                                 |
| <a href="#">PMID:36039087</a> | Model organism (rodent)              | <b>Supports (downstream)</b> | Antidepressant excludes/degrades FOXO1              | SIRT1 drives FOXO1 cytoplasmic translocation + degradation                                                           |
| <a href="#">PMID:41249439</a> | Model organism (rodent)              | <b>Supports (downstream)</b> | AKT inhibition → FoxO3a nuclear → Puma apoptosis    | NDRG2→PP2A→AKT ↓ →FoxO3a→Puma drives astrocyte apoptosis & depression                                                |
| <a href="#">PMID:37173696</a> | Model organism (rodent)              | <b>Supports (downstream)</b> | TLR4/PI3K/AKT/FOXO1 in neuroinflammatory depression | Antidepressant suppresses FOXO1 nuclear export                                                                       |
| <a href="#">PMID:40237232</a> | Model organism (rodent)              | <b>Supports (downstream)</b> | SIRT1/FOXO1 microglial axis                         | Urolithin B restores SIRT1/FOXO1 antidepressant                                                                      |
| <a href="#">PMID:41622191</a> |                                      | <b>Qualifies / conflicts</b> | FoxO3a direction in dentate gyrus                   | AAV-FoxO3a overexpression alleviates depression-like behaviour                                                       |

| Citation                                                      | Evidence type           | Role                         | Mechanistic claim tested                           | Key finding                                                                     |
|---------------------------------------------------------------|-------------------------|------------------------------|----------------------------------------------------|---------------------------------------------------------------------------------|
|                                                               | Model organism (rodent) |                              |                                                    |                                                                                 |
| <a href="#">PMID:42632434</a>                                 | Human clinical          | <b>Refutes (biomarker)</b>   | Plasma FOXO3 tracks depressive symptoms            | After adjustment, FOXO3 unassociated with depressive symptoms (P=0.2)           |
| Open Targets v26.9 (computational)                            | Computational           | <b>Qualifies</b>             | Which FOXO/insulin genes have MDD genetic evidence | Only FOXO1 has a credible set (L2 0.469); INSR/IGF1R/SGK1/AKT1/FOXO3/4/6 none   |
| <a href="#">PMID:34782712</a> (GCST90096932)                  | Human genetic           | <b>Qualifies</b>             | Source of FOXO1 signal                             | rs180828263 from cognition×MDD interaction GWAS, not primary MDD GWAS           |
| <a href="#">PMID:40595333</a>                                 | Human genetic (MR)      | <b>Qualifies / competing</b> | Insulin pathway–MDD causality                      | Bidirectional MDD↔T2D; obesity-mediated insulin resistance                      |
| <a href="#">PMID:41555281</a>                                 | Human genetic           | <b>Qualifies</b>             | Shared MDD–MetS architecture                       | Bidirectional genetic correlation, glucose/insulin pathways                     |
| <a href="#">PMID:41780463</a>                                 | Review (genomic)        | <b>Qualifies</b>             | MDD→MetS causal direction                          | Unidirectional MDD→MetS; shared glucose/lipid/inflammation pathways             |
| <a href="#">PMID:35854107</a> / <a href="#">PMID:42538881</a> | Review (umbrella)       | <b>Refutes (upstream)</b>    | Is MDD associated with low serotonin?              | No consistent evidence for lower serotonin concentration/activity               |
| <a href="#">PMID:36356194</a>                                 | Review / commentary     | <b>Qualifies (upstream)</b>  | Serotonin hypofunction scope                       | Characteristic of only a depressed subgroup (agitated/anxious, SSRI-responsive) |
| <a href="#">PMID:28088287</a>                                 |                         |                              |                                                    |                                                                                 |

| Citation | Evidence type           | Role                       | Mechanistic claim tested   | Key finding                                                                          |
|----------|-------------------------|----------------------------|----------------------------|--------------------------------------------------------------------------------------|
|          | Model organism (rodent) | <b>Qualifies (context)</b> | IGF/insulin–monoamine link | Igfbp3-null mice: ↓ IGF-1, ↓ Akt/ERK1/2, ↓ serotonin & dopamine, behavior impairment |

## Data and Tool Use

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### Datasets, databases, and APIs

| Resource                          | Accession / URI                                                            | Version / snapshot | Retrieval date | Access status                 | Query / filter                                                                                                                                                 | Resolution / relevance                    |
|-----------------------------------|----------------------------------------------------------------------------|--------------------|----------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Open Targets Platform GraphQL API | https://api.platform.opentargets.org/api/v4/graphql; disease MONDO_0002009 | v26.9.0            | 2026-10-09     | Accessed (succeeded)          | Target–disease association scores + genetic evidence for FOXO1/3/4/6, INSR, IGF1R, SGK1, AKT1, SLC6A4, TPH2; top-30 MDD targets; FOXO1 credible-set resolution | Resolution direct relevance               |
| Open Targets credible set (FOXO1) | variant rs180828263; study GCST90096932                                    | v26.9.0            | 2026-10-09     | Accessed (succeeded)          | L2G, PICS fine-mapping, parent study metadata                                                                                                                  | Resolution relevant but subtle conditions |
| PubMed / NCBI E-utilities         | —                                                                          | live               | 2026-10-08/09  | Searched (34 papers reviewed) | See negative-search log below                                                                                                                                  | Relevance                                 |
| GWAS Catalog (via Open Targets)   | GCST90096932                                                               | —                  | 2026-10-09     | Cited via Open Targets        | Parent-study identity of FOXO1 signal                                                                                                                          | Resolution relevant                       |
| GenCC / ClinGen                   | —                                                                          | —                  | —              | Not queried (unverified)      | Gene–disease validity for                                                                                                                                      | Flag and unverifiable absence             |

| Resource | Accession / URI | Version / snapshot | Retrieval date | Access status | Query / filter | Resolution |
|----------|-----------------|--------------------|----------------|---------------|----------------|------------|
|----------|-----------------|--------------------|----------------|---------------|----------------|------------|

FOXO1/3/4  
in MDD

## Negative / null searches (search log)

| Query                                                             | Database           | Date       | Result                                                            |
|-------------------------------------------------------------------|--------------------|------------|-------------------------------------------------------------------|
| "serotonin 5-HT receptor FOXO transcription factor brain neurons" | PubMed             | 2026-10-09 | Only a <i>C. elegans</i> paper; no mammalian serotonin→FOXO study |
| "fluoxetine FOXO3a hippocampus phosphorylation"                   | PubMed             | 2026-10-09 | 0 results                                                         |
| "fluoxetine Akt FOXO neuronal"                                    | PubMed             | 2026-10-09 | 0 results                                                         |
| INSR / IGF1R / SGK1 / AKT1 MDD GWAS credible-set evidence         | Open Targets v26.9 | 2026-10-09 | 0 genetic rows for each                                           |
| FOXO3 / FOXO4 / FOXO6 MDD genetic credible-set/burden             | Open Targets v26.9 | 2026-10-09 | FOXO3 literature-only; FOXO4/6 no association                     |

## Analyses attempted (input → method → output)

| Analysis                                       | Trace                                                                                                                                                                                                        | Environment                                        | Outcome          | Limitation                                                     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------|----------------------------------------------------------------|
| FOXO/<br>insulin<br>MDD<br>association<br>scan | Open Targets GraphQL (MONDO_0002009) →<br>GraphQL queries →<br><code>openscientist_artifacts/data/<br/>opentargets_foxo_mdd_associations.csv</code><br>+<br><code>queries/opentargets_queries.graphql</code> | Python<br>(requests),<br>Open Targets<br>v26.9 API | <b>Succeeded</b> | API scores<br>are<br>aggregate;<br>no raw<br>patient<br>data   |
| FOXO1<br>credible-<br>set<br>resolution        | Open Targets GraphQL →<br><code>openscientist_artifacts/data/<br/>foxo1_mdd_crediblesets.json</code>                                                                                                         | Python, v26.9<br>API                               | <b>Succeeded</b> | PICS<br>"reported<br>top hit",<br>not full<br>fine-<br>mapping |
| MDD top-<br>target<br>context                  | Open Targets GraphQL (6,780 targets) →<br><code>openscientist_artifacts/data/<br/>mdd_top_targets_context.csv</code>                                                                                         | Python, v26.9<br>API                               | <b>Succeeded</b> | Overall<br>score is a<br>composite<br>heuristic                |

No patient-level, omics, or controlled-access dataset was accessed; no such analysis was run. No structural-biology (Phenix) analysis was applicable to this literature/genetics hypothesis. **No analysis was silently substituted for another:** where a direct serotonin→FOXO literature link was sought and not found, that is reported above as a negative search, not as a completed analysis.

## Mechanistic Causal Chain — where literature is strong, inferred, or missing

| Causal step                                   | Edge                                    | Status                                                                                                  |
|-----------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| Serotonergic tone (upstream trigger)          | —                                       | <b>Contested</b> in humans (F007); subtype-restricted at best                                           |
| 5-HT → insulin/IGF-1 receptor signalling      | 5-HT → DAF-2/INSR                       | <b>Strong in worm</b> (F001); <b>no human genetic support</b> for INSR/IGF1R (F004)                     |
| Insulin/IGF-1 → FOXO nuclear exclusion        | DAF-2/AKT → DAF-16/FOXO                 | <b>Strong in worm; supported in mammal</b> (AKT→FOXO) but <b>directionally conflicted</b> (F002/F003)   |
| FOXO de-repression → stress/apoptotic program | FOXO → Bim/Puma/inflammation            | <b>Supported in mammal</b> (F002)                                                                       |
| Stress program → depression phenotype         | → MDD                                   | <b>Inferred</b> ; human FOXO3 biomarker negative (F003); FOXO1 subtype-conditioned (F006)               |
| Drugs keep FOXO cytoplasmic (therapeutic)     | antidepressant/lithium/clozapine → FOXO | <b>Strong in worm; echoed in mammal</b> via AKT/SIRT1/β-arrestin, <b>not demonstrably via serotonin</b> |

The single missing causal step that defines the hypothesis — a demonstrated serotonin→FOXO edge in a mammal — was not found. This is the decisive gap for curation.

## Limitations and Knowledge Gaps

1. **No mammalian serotonin→FOXO edge (critical gap).** Scope: the defining link of the hypothesis. Why it matters: without it, the mammalian AKT→FOXO data support a *generic* stress hub, not a serotonin-restrained one. Checked: three targeted PubMed searches (2026-10-09) returned nothing usable. Resolution: a direct perturbation — e.g., serotonin depletion (PCPA) or 5-HT receptor agonism/antagonism in rodent hippocampus with FOXO nuclear-localisation readout, ± fluoxetine rescue.

2. **Wrong-family-member mismatch.** Mammalian mechanism centres FOXO3a (apoptosis/Puma/Bim); the only human genetic signal is FOXO1; FOXO3 is literature-only with a negative human biomarker (F003/F004). Resolution: human brain eQTL/expression and genetic fine-mapping distinguishing FOXO1 vs FOXO3 in MDD cohorts.
  3. **Insulin-receptor arm has no human MDD genetic support.** INSR/IGF1R/SGK1/AKT1 all have 0 genetic credible-set rows in Open Targets v26.9 (F004). The human insulin–MDD link that exists is a metabolic comorbidity (F005), a different edge. Resolution: colocalisation of MDD GWAS with insulin-pathway eQTLs in brain tissue.
  4. **Directional conflict in rodents (unresolved).** Nuclear FoxO3a is pro-depressive in astrocytes ([PMID:41249439](#)) yet protective in dentate-gyrus neurons ([PMID:41622191](#)). Scope: whether FOXO is net harmful or protective depends on cell type/region. Resolution: cell-type-specific conditional FOXO knockout/overexpression with matched behavioural batteries.
  5. **Contested upstream premise.** The "loss of serotonin" node is not consistently supported in humans ([PMID:35854107](#)); it may hold only for an agitated/anxious SSRI-responsive subtype ([PMID:36356194](#)). Resolution: subtype-stratified serotonergic markers paired with FOXO readouts.
  6. **FOXO1 signal is subtype-conditioned and weak.** rs180828263 arises from a cognition×MDD interaction GWAS (GCST90096932), PICS "reported top hit," L2G 0.469 (F006). Resolution: replication in independent primary MDD GWAS and in cognitive-subtype-stratified cohorts.
  7. **Source/dataset absences checked.** No clinical/known-drug evidence tying FOXO genes to MDD therapeutics in Open Targets v26.9 (2026-10-09). GenCC and ClinGen were **not** separately queried and must be labeled **unverified** rather than claimed absent until a dated query is logged.
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## Alternative Models

| Model                                                                                                                          | Relationship to seed                             | Summary                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monoamine / synaptic genetic core</b> (DRD2, HTR2A, GRIN2A, GABRA1, SLC6A4, HTR1A)                                          | <b>Competing / more parsimonious</b>             | Dominates MDD's human genetic architecture (F008); explains antidepressant pharmacology without a FOXO node.                                         |
| <b>AKT/FOXO as a convergent downstream stress hub</b>                                                                          | <b>Downstream consequence</b>                    | FOXO engaged by SIRT1, PI3K/AKT, TLR4, $\beta$ -arrestin/SGK — a shared effector of many antidepressant mechanisms, not serotonin-specific (F002).   |
| <b>Insulin-resistance / metabolic-comorbidity model</b>                                                                        | <b>Parallel mechanism / upstream in a subset</b> | Bidirectional MDD $\leftrightarrow$ T2D, obesity-mediated (F005); implicates insulin pathway but via metabolic route, not 5-HT gating.               |
| <b>Neuroinflammation (TLR4/NF-<math>\kappa</math>B, microglia)</b>                                                             | <b>Parallel / upstream</b>                       | Drives FOXO1 and depressive behaviour ( <a href="#">PMID:37173696</a> , <a href="#">PMID:40237232</a> ); FOXO is effector, inflammation the trigger. |
| <b>Neurotrophic/BDNF-neurogenesis model</b>                                                                                    | <b>Parallel / downstream</b>                     | CUMS reduces BDNF/pCREB alongside FOXO changes (F002); FOXO3a protective via neuroplasticity in DG (F003).                                           |
| <b>IGF/BDNF/serotonin "triumvirate" of stress signalling</b> ( <a href="#">PMID:15541711</a> , <a href="#">PMID:28088287</a> ) | <b>Upstream context</b>                          | Couples IGF signalling to monoamines generically, consistent with but broader than the serotonin $\rightarrow$ FOXO edge.                            |

## Discriminating Tests

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1. **Direct serotonin→FOXO perturbation in rodent brain (highest priority).** Deplete serotonin (PCPA) or manipulate specific 5-HT receptors in hippocampus/cortex; read out FOXO1 and FOXO3a nuclear:cytoplasmic ratio by quantitative immunofluorescence; test fluoxetine rescue. *Expected if hypothesis true:* serotonin loss drives FOXO nuclear accumulation, reversed by fluoxetine, and blocked by constitutively active AKT.
  2. **Cell-type-specific FOXO manipulation with behaviour.** Conditional FOXO1 vs FOXO3a knockout/overexpression in astrocytes vs dentate-gyrus neurons under CUS. *Expected:* resolves the pro- vs anti-depressive directional conflict and identifies the disease-relevant family member.
  3. **Human brain colocalisation.** Colocalise MDD GWAS signals (primary case/control, and cognition-stratified) with brain eQTLs for FOXO1, FOXO3, INSR, IGF1R, SGK1. *Expected if hypothesis true:* shared causal variants at insulin-pathway/FOXO loci; *expected given current data:* little/no colocalisation except possibly FOXO1 in the cognitive subtype.
  4. **Subtype-stratified biomarker study.** In SSRI-responsive agitated/anxious MDD vs anhedonic/anergic MDD, measure CSF/plasma serotonergic markers and peripheral/brain FOXO activity. *Expected if subtype-restricted:* serotonin-FOXO coupling appears only in the SSRI-responsive subtype.
  5. **SSRI longitudinal FOXO readout.** Measure FOXO target-gene signatures (Bim, Puma, catalase, SOD) before/after SSRI in responders vs non-responders. *Expected if hypothesis true:* response tracks FOXO target-program suppression.
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## Curation Leads (require curator verification)

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- **Status / node decision:** Do **not** promote an insulin/FOXO pathophysiology node for MDD-at-large. Curate as a **HUMAN\_MODEL\_MISMATCH** discussion attached to **Monoamine Deficiency**, retaining an **EMERGING, subtype-restricted FOXO1 lead** (cognitive subtype).
- **Candidate evidence references & snippets to verify:**
- [PMID:42632434](#) — "After adjustment, FOXO3 remained unassociated with depressive symptoms ( $P = 0.232$ ).\" (human negative biomarker)

- [PMID:41249439](#) — *"promotes the nuclear translocation of FoxO3a, leading to the transcriptional activation of the pro-apoptotic factor Puma"* (mammalian pro-depressive FOXO3a)
  - [PMID:41622191](#) — FoxO3a dentate-gyrus overexpression *alleviates* depression (directional conflict)
  - [PMID:36039087](#) — *"CSS serum could also promote FOXO1 translocation to the cytoplasm through SIRT1 signaling, which triggered FOXO1 protein degradation."* (antidepressant excludes FOXO1)
  - [PMID:35854107](#) — umbrella review contesting the serotonin-deficiency premise
  - [PMID:36356194](#) — *"The decreased activity of serotonin ... is characteristic of only a subgroup of depressed subjects"*
  - [PMID:40595333](#) — MR bidirectional MDD↔T2D (insulin as metabolic comorbidity)
  - **Candidate pathophysiology edges (as discussion, not confirmed):** AKT → FOXO (nuclear exclusion) [mammalian, directionally conflicted]; antidepressant → FOXO cytoplasmic sequestration [mammalian]; MDD ↔ insulin resistance [human MR, metabolic arm]. Mark the serotonin → FOXO edge as **unconfirmed in mammals**.
  - **Candidate ontology terms:** FOXO1 (HGNC:3819), FOXO3 (HGNC:3821); biological processes — forkhead-box transcription-factor stress response, insulin/IGF-1 receptor signalling (DAF-2 ortholog INSR/IGF1R), intracellular protein localisation (nuclear export); cell types — hippocampal dentate-gyrus granule neuron, astrocyte, microglia.
  - **Candidate subtype restriction:** Restrict any serotonin-dependent FOXO claim to a **cognitive / agitated-anxious SSRI-responsive MDD subtype**, reflecting both the FOXO1 interaction-GWAS source (GCST90096932) and the serotonin-subtype commentary.
  - **Candidate knowledge\_gaps / discussion prompts:** (1) No demonstrated mammalian serotonin→FOXO edge; (2) wrong-FOXO-family-member mismatch (mammal FOXO3a vs human-genetic FOXO1); (3) no human MDD genetic support for INSR/IGF1R/SGK1/AKT1; (4) unresolved pro- vs anti-depressive FOXO directionality; (5) GenCC/ClinGen not yet queried — flag as **unverified** absence pending a dated search.
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## Provider Artifact Bundle

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Under

`kb/hypotheses/Major_Depressive_Disorder/`

`serotonin_foxo_stress_transcription_model/openscientist_artifacts/`:

- `data/opentargets_foxo_mdd_associations.csv` — FOXO/insulin gene MDD association & genetic-evidence scores (Open Targets v26.9, 2026-10-09)
- `data/foxo1_mdd_crediblesets.json` — FOXO1 credible-set resolution (rs180828263 → GCST90096932)
- `data/mdd_top_targets_context.csv` — top-30 MDD targets context (DRD2/HTR2A/GRIN2A/GABRA1 ...)
- `queries/opentargets_queries.graphql` — exact GraphQL queries used

The bundle is non-empty because computational Open Targets analyses were actually run. No large raw downloads, controlled data, or credentials are bundled; the external Open Targets snapshot is identified by version (v26.9.0) and disease ID (MONDO\_0002009). A canonical `MANIFEST.yaml` with checksums and replay verification should be finalised by the curator if not already present. No patient-level or omics dataset was accessed, and none is claimed.

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## Conclusion

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The Serotonin-Restrained FOXO Stress Transcription Model is a mechanistically elegant chain that is **fully supported only in *C. elegans***. In mammals, its downstream AKT→FOXO→stress-program arm is independently and repeatedly supported, including reversal by antidepressants, but its defining serotonin→FOXO edge is untested, its FOXO directionality is conflicted, its human FOXO3 biomarker is negative, and human genetics nominate only a weak, subtype-conditioned FOXO1 signal while giving no support to the insulin-receptor arm. The human insulin–MDD link that does exist is a parallel metabolic comorbidity, and the upstream serotonin-deficiency premise is itself contested. **Recommended disposition: HUMAN\_MODEL\_MISMATCH discussion attached to Monoamine Deficiency, with a subtype-restricted EMERGING FOXO1 lead — not a promoted primary pathophysiology node.**

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